Policy-BT: comparison contract and tournament mathematics

Implementation-level view • task criteria remain fixed while artifact-schema adapters can expand.

A Evidence and task contract

Owned workspace snapshot
Original artifacts remain untouched
Required .sh reproduction contract

Re-execution gate
Clean venv; move old outputs aside
Rerun optional; contract required

Fail → admission verdict; reward 0

pass

LLM reconnaissance
Goal + inventory + inspected previews
Fix criteria, stage order and value types

Schema-specific adapters
Artifacts → values + path evidence

B Code first; images on ties

One cached comparison policy
Validity / output conformance → quality
Earliest decisive criterion wins
Unknown values do not imply failure

Restricted interpreter + bounded repairs

Check both orders, both self-ties
and repeated forward execution.

Code tie only

Relevant visual evidence
Recon-selected images from both sides
Vision model uses the same goal + criteria
Accept only order-consistent outcomes
Unresolved → pair stays tied

Code decides

Verdict · stage · evidence · feedback

C Persistent task tournament

Every unordered pair
Artifact + admission verdicts
Pair ledger keyed by bundle version
Bundle change → recompute all pairs

Regularized strength estimation
All pairs, including admission verdicts
Fit latent θ; publish eligible σ(θ)

Refresh all current rewards
Selection uses the refreshed ratings

Separate textual gradient
Order losses first, then earliest stage
Retain diagnoses; omit scores / code

BRADLEY–TERRY MODEL

$$p_{ij} = \sigma(\theta_i - \theta_j)$$

y_{ij} = 1 (win), ½ (tie), 0 (loss)

REGULARIZED OBJECTIVE

$$\min_{\theta}: -\Sigma\,[y_{ij}\log p_{ij} + (1-y_{ij})\log(1-p_{ij})] + \tfrac{1}{2}\,\Sigma\,\theta_i^2$$

Unit Gaussian prior • $r_i = \sigma(\theta_i)$ versus a neutral-strength opponent; ineligible $r_i = 0$.

No-pair eligible baseline: 0.5. Instrument error: unscored, not reward 0. Reruns are trusted local shell execution, not an OS sandbox.